

**A**  
**Internship Report**  
**On**  
**“ SnapText OCR ”**

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**Submitted to**



**Information Technology Department**

**DR. S & S. S. GHANDHY COLLEGE OF ENGINEERING &  
TECHNOLOGY, SURAT**

**GUJARAT TECHNOLOGY UNIVERSITY**

**August 2023**



## CERTIFICATE

This is to certify that work embodied in this report entitled “**SnapText OCR**” was carried out by **Mr. Krish Ujeniya** at **Dr. S & S. S. Ghandhy College of Engineering & Technology, Surat.**

For partial successful completion of **Summer Internship - II (4351605)** of **Semester-5 Diploma engineering in Information Technology** to be awarded by Gujarat Technological University. This internship work has been carried out under my supervision and is to the satisfaction of the department.

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Place: Surat

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## DECLARATION OF ORIGINALITY

We hereby certify that I am the sole author of this internship and that neither any part of this work nor the whole work has been submitted for a degree to any other University or Institution.

We certify that, to the best of our knowledge, our work does not infringe upon anyone's copyright nor violate any proprietary rights and that any ideas, techniques, quotations, or any other material from the work of other people included in our report or otherwise, are fully acknowledged in the accordance with the standard referencing practices.

I declare that this is a true copy of our report, including any final revisions, as approved by our supervisor.

**Date:**

**Place:** Surat

**Yours Sincerely,**

Krish Ujeniya

(216120316028)

## ACKNOWLEDGEMENT

We are so glad to present our internship report on **‘SnapText OCR’**, which is prepared as a part of our fifth semester Summer Internship-2 in Diploma Engineering & Technology. This work is the outcome of guidance provided by the dedicated Information Technology, Dr. S. & S. S. Ghandhy College of Engineering & Technology.

With a profound sense of indebtedness, we use this opportunity to express our sincere gratitude to Prof. Manish D Patel, H.O.D. of the Information Technology department for inspiring and permission to carry out our internship.

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Last but not least, thanks to **Dr. S. & S. S. Ghandhy College of Engineering & Technology** for providing us the platform to represent the internship.

**Yours Sincerely,**

Krish Ujeniya

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## Abstract

The **SnapText OCR** is the web portal that provides management of ration in online mode. SnapText OCR provides a multi-mode payment system, an email notifying service for users to notify that ration is available for user. SnapText OCR overcomes the flaws of the existing manual system by storing records online.

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# Chapter-1 Introduction

## 1.1 Project Defination

The project, titled "SnapText," aims to develop a mobile application that leverages Google ML Kit and Flutter technologies to extract text from images captured by users.

The primary goal of SnapText is to provide users with a convenient and efficient way to digitize printed or handwritten text, making it editable and searchable. The application targets a wide range of users, including students, professionals, and anyone who needs to quickly convert physical text into digital format.

### ❖ Scope of the Project

The scope of the SnapText project encompasses the following key features and functionalities:

- **Image Capture:** Users can capture images of printed or handwritten text using the device's camera or select images from their gallery.
- **Text Recognition:** SnapText employs OCR technology to recognize and extract text from the captured images accurately.
- **User Authentication:** The application includes user authentication to ensure data privacy and access control.
- **User History:** Recognized text is stored in a user-specific history section, allowing users to revisit and edit previous scans.

- **QR Code Generation:** SnapText also provides a QR code generation feature, enabling users to generate QR codes from recognized text.
- **Multi-Platform Support:** The application is designed to run on both Android and iOS platforms, ensuring a broad user base.

## 1.2 Company Profile

### ❖ About Prelax Infotech:

Prelax Infotech is a certified company established in 2013. We offer services in website designing, mobile app development, digital marketing, game development, and more. Our core values are collaboration, integrity, and accountability, which are deeply ingrained in our work.

### ❖ Company Details:

**Email:** [info@prelax.in](mailto:info@prelax.in)

**Instagram:** [https://www.instagram.com/prelax\\_infotech/](https://www.instagram.com/prelax_infotech/)

**Linkedin:** <https://in.linkedin.com/in/prelax-infotech-hr-4413a3184>

**Website:** <https://prelax.in>

**Address:** 615 to 618, Infinity Tower, Lal Darwaja Station Rd, beside Ayurvedic Collage, near the railway station, Lal Darwaja, Surat, Gujarat 395003

## **Chapter-2 Requirement Analysis**

### **2.1 Software Requirement Specification**

#### **2.1.1. Introduction**

##### **a. Purpose**

The purpose of this document is to provide a detailed description of the requirements for the SnapText Flutter application. This document outlines the functionality, constraints, and features of the application.

##### **b. Scope**

The SnapText application is designed to assist users in capturing images, recognizing text within images, generating QR codes, and managing their recognized text history. It includes user authentication for secure access to the application.

#### **2.2.2. Functional Requirements**

##### **2.2.2.1 User Authentication**

###### **a. User Registration**

Users can register for a new account by providing their email address and password.

Passwords must meet specified security requirements, including minimum length and complexity.

Users receive a confirmation email for account verification.

**b. User Login**

Registered users can log in using their email and password.

Authentication should be secure, and user sessions should be maintained.

**c. Password Reset**

Users can request a password reset email if they forget their password.

A reset link is sent to the user's email for password recovery.

**d. User Logout**

Logged-in users can log out of their accounts.

**2.2.2.2 Image Capture and Processing****a. Image Capture**

Users can capture images from the device's camera or select images from the device's gallery.

Captured images should be stored temporarily for processing.

**b. Text Recognition**

Users can initiate text recognition on captured images.

Recognized text from images is displayed to the user.

Recognized text is stored in the user's history.

**2.3 QR Code Generation****a. Text Input**

Users can enter text that they want to convert into a QR code.

The input text should be validated for length and content.

#### **b. QR Code Generation**

Users can generate QR codes from the entered text.

Generated QR codes are displayed to the user for sharing or saving.

### **2.2.2.4 History Management**

#### **a. View History**

Users can view a list of recognized text items from their history.

Each item in the history is clickable to view the details.

#### **b. History Details**

Users can view the details of recognized text items, including the recognized text and timestamp.

Options to edit, copy to clipboard, search in a browser, and translate are available.

### **2.2.3. Non-Functional Requirements**

#### **2.2.3.1 Performance**

The application should provide a responsive and smooth user experience.

Image processing and text recognition should occur within a reasonable timeframe.

#### **2.2.3.2 Security**

User data and credentials should be securely stored and transmitted using encryption.

Authentication mechanisms should be robust and resistant to common security threats.

### **2.2.3.3 Usability**

The user interface should be intuitive and user-friendly.

Adequate error messages and feedback should be provided to users.

### **2.2.3.4 Compatibility**

The application should be compatible with Android and iOS platforms.

It should work on a variety of device screen sizes and orientations.

## Chapter-3 System Design

### 3.1 E-R Diagram

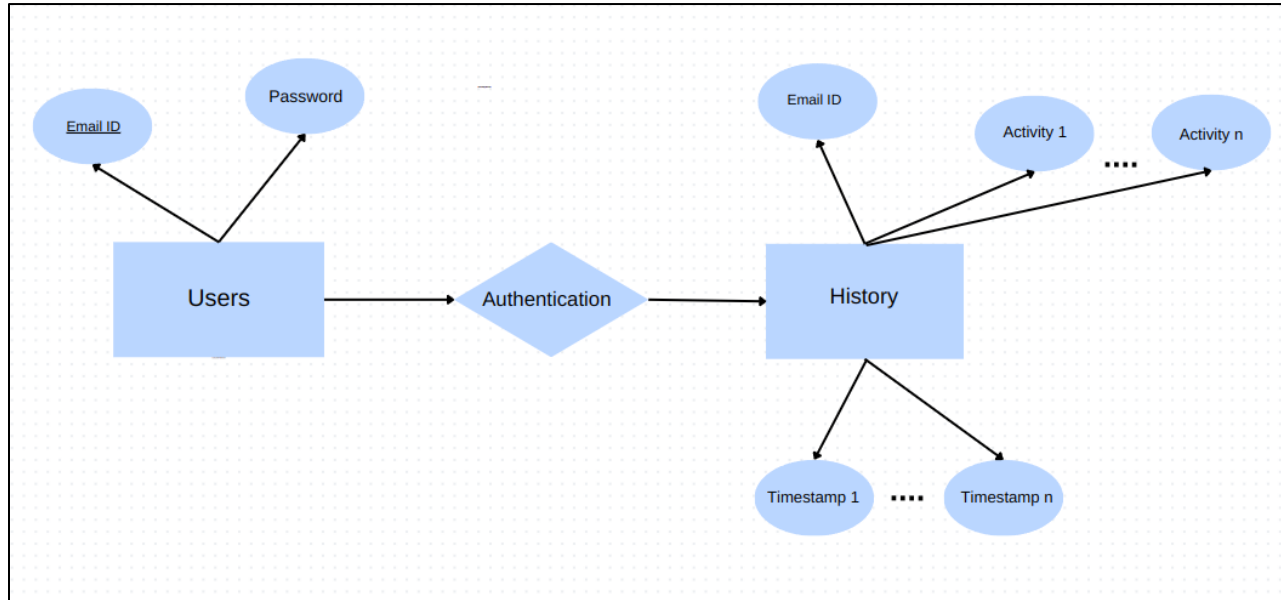


Fig 3.1 Proposed systems E-R Diagram

#### Entities:

- Users
  - Email ID
  - Password
- History
  - Email ID
  - Activity 1 ... Activity n
  - Timestamp 1 ... Timestamp n

#### -Relationship:

- Authentication

## 3.2 Data Flow Diagram

### 3.2.1 Level 0 :

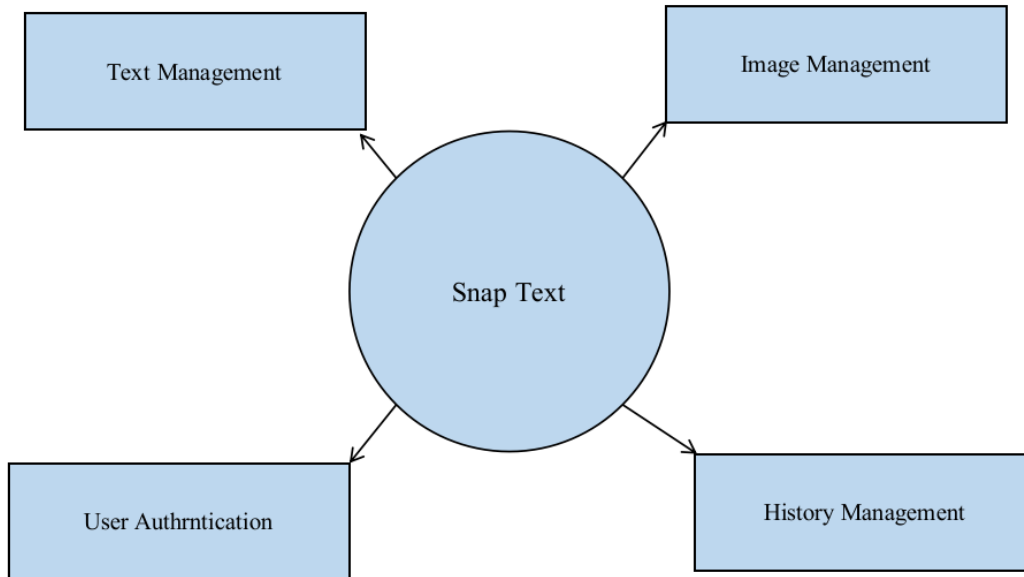


Fig 3.2 Context Level Diagram

### 3.2.2 Level 1 :

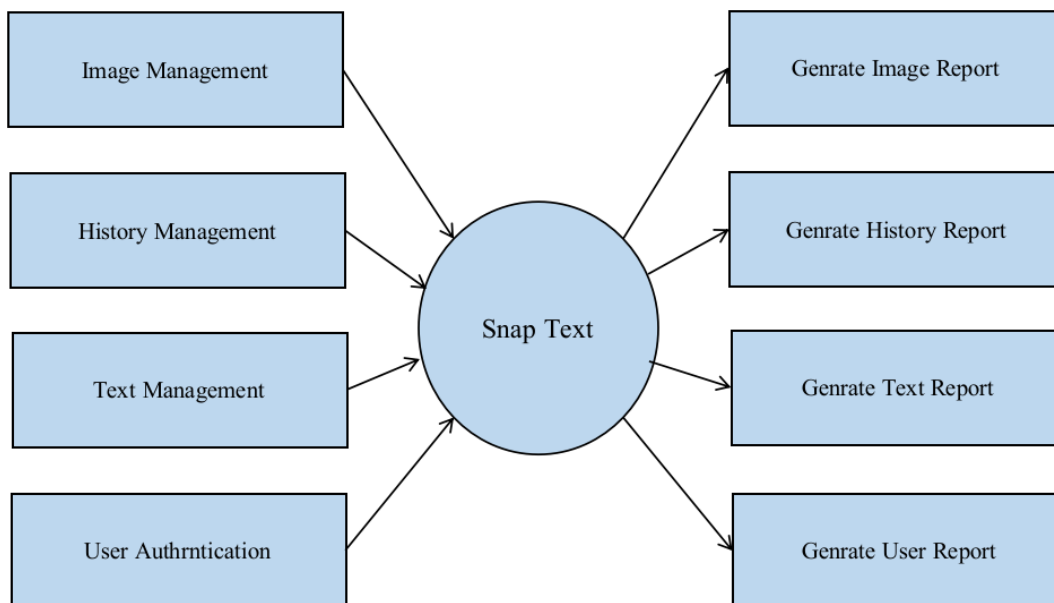


Fig 3.3 Level-1 Dataflow Diagram



### 3.2.3 Level 2 :

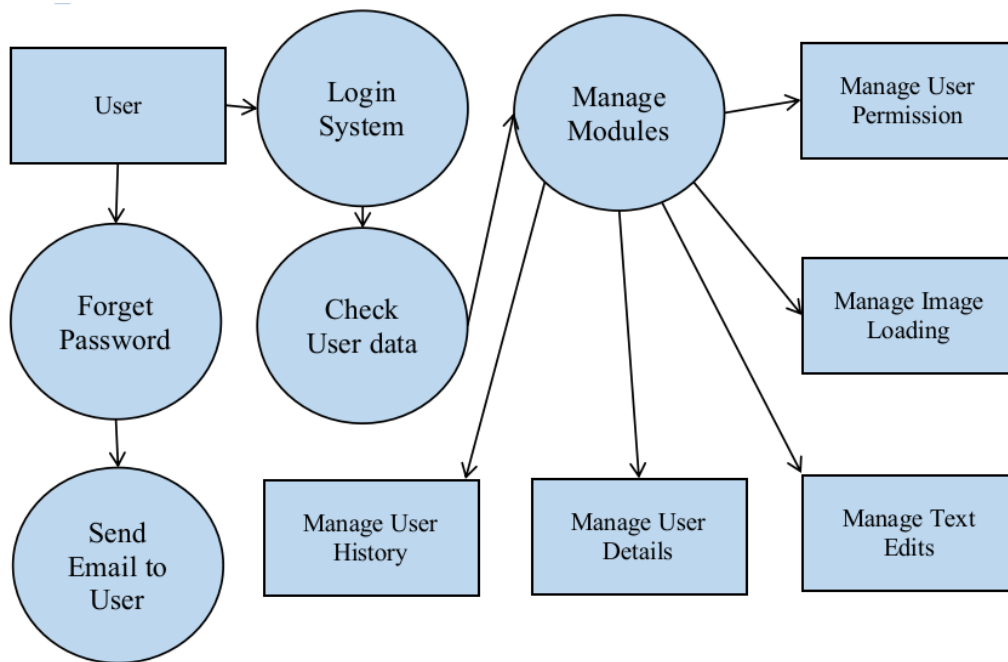


Fig 3.4 Level-2 Dataflow Diagram

### 3.3 Data Dictionary

#### ❖ Users:

| Field Name | Description                 | Data Type | Length | Constraints             | Example                                      |
|------------|-----------------------------|-----------|--------|-------------------------|----------------------------------------------|
| Email ID   | Unique identifier for users | Text      | 100    | Primary Key, Unique Key | <a href="mailto:abc@xyz.com">abc@xyz.com</a> |
| Password   | User's password             | Text      | 100    |                         | abc@102                                      |

Fig 3.1 Data Dictionary of user

❖ **History:**

| Field Name      | Description                     | Data Type | Length | Constraints             | Example                                      |
|-----------------|---------------------------------|-----------|--------|-------------------------|----------------------------------------------|
| Email ID        | Unique identifier for users     | Text      | 100    | Refrence Key,Unique Key | <a href="mailto:abc@xyz.com">abc@xyz.com</a> |
| Activity 1 .. n | User's Recognized text history  | Text      | 255    |                         | snapText                                     |
| TimeStamp       | User when reconginize text show | DateTime  |        |                         | 10-09-2023<br>15.09.88                       |

Fig 3.2 Data Dictionary of history

## Chapter 4: GUI/Application Interface

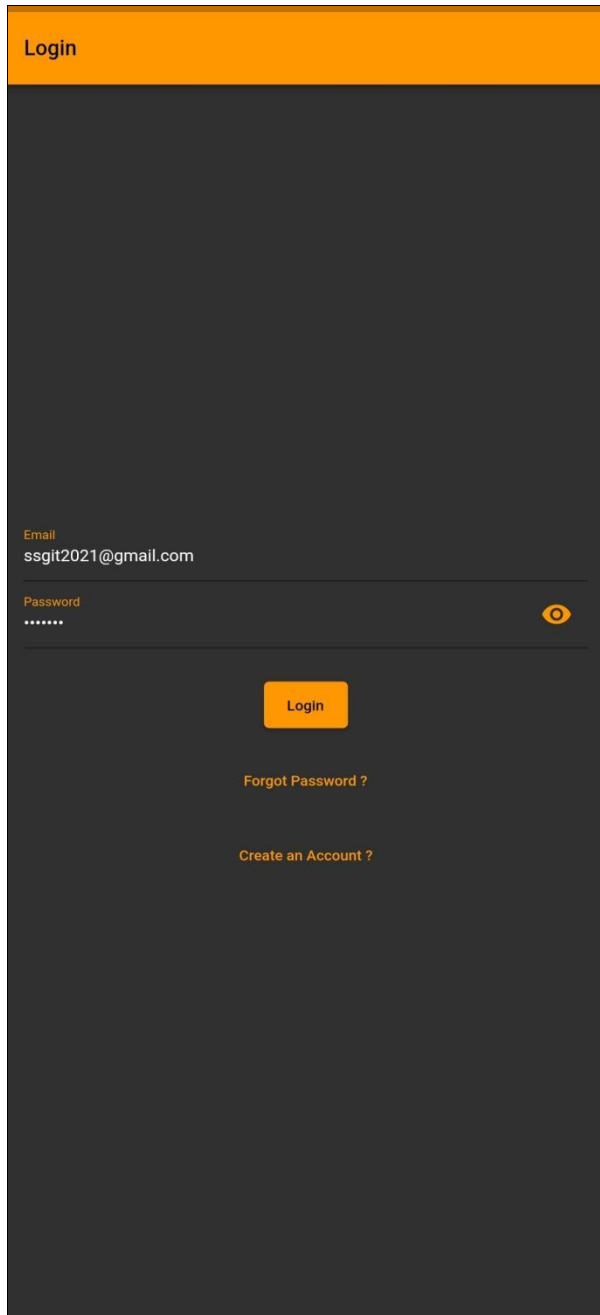
### 4.1 User Side:



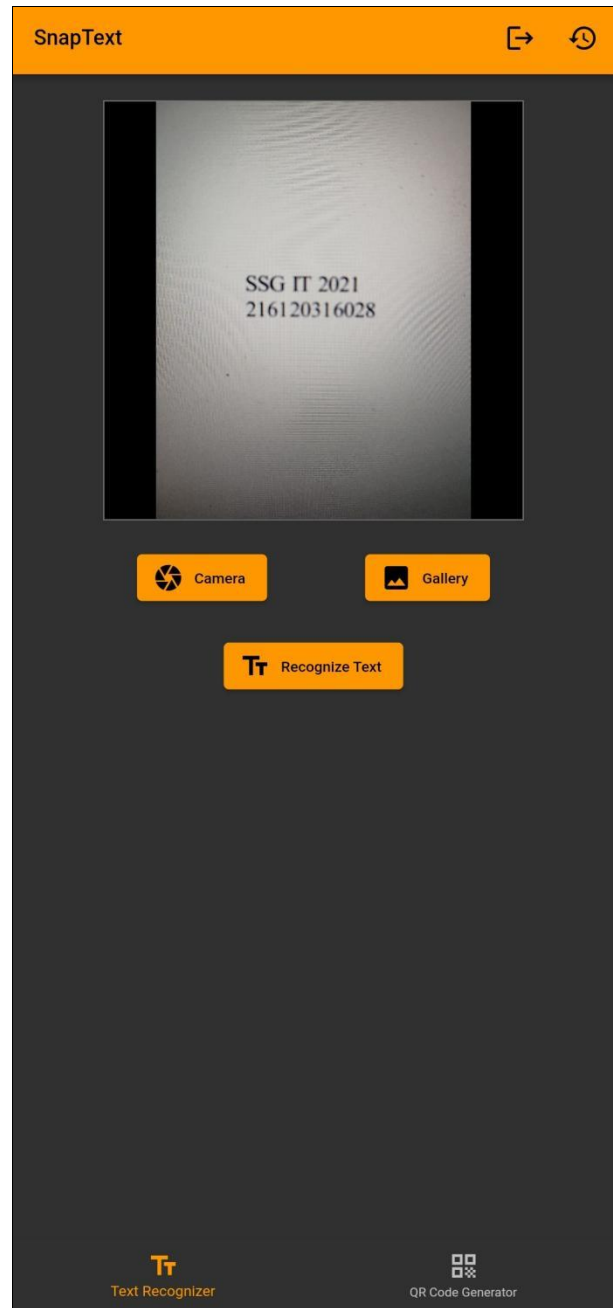
4.1 SplashScreen

The image shows a registration screen with a dark grey background. At the top, there is an orange header bar with the word "Registration" in white. Below the header, there are three input fields. The first field is labeled "Email" and contains the text "ssgit2021@gmail.com". The second field is labeled "Password" and contains six dots. To the right of the password field is an orange eye icon. The third field is labeled "Confirm Password" and contains six dots. To the right of the confirm password field is an orange eye icon. Below the input fields, there is an orange button with the word "Register" in white. At the bottom, there is a link that says "Already Have Account ?" in orange.

4.2 RegistrationScreen



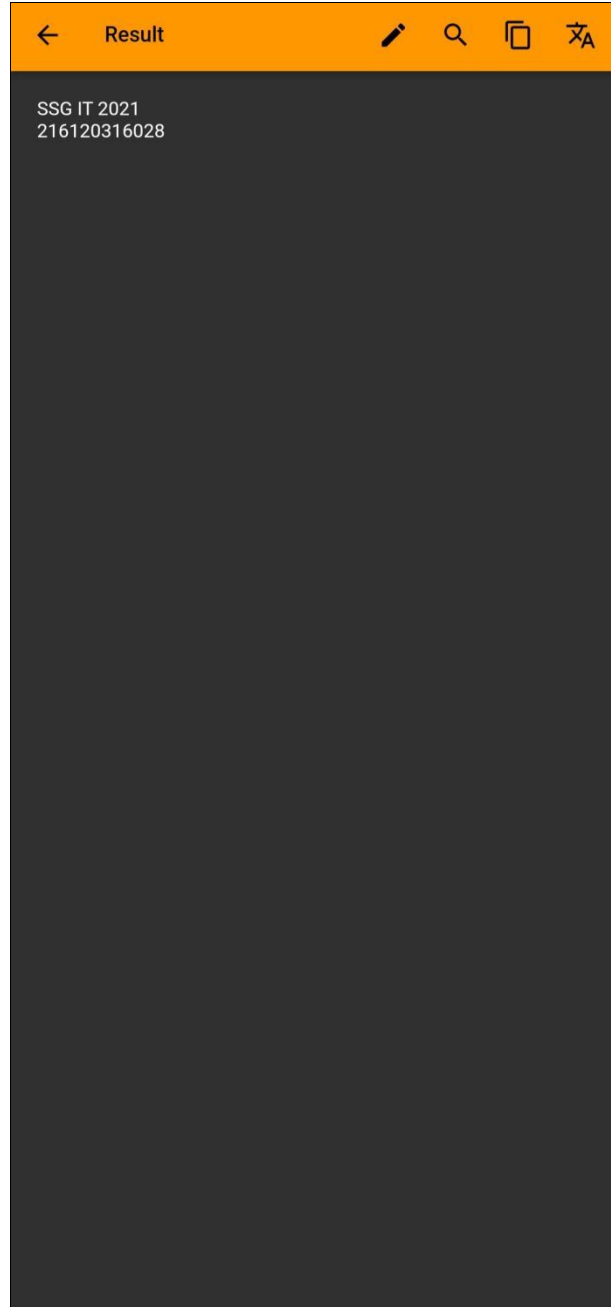
4.3 LoginScreen



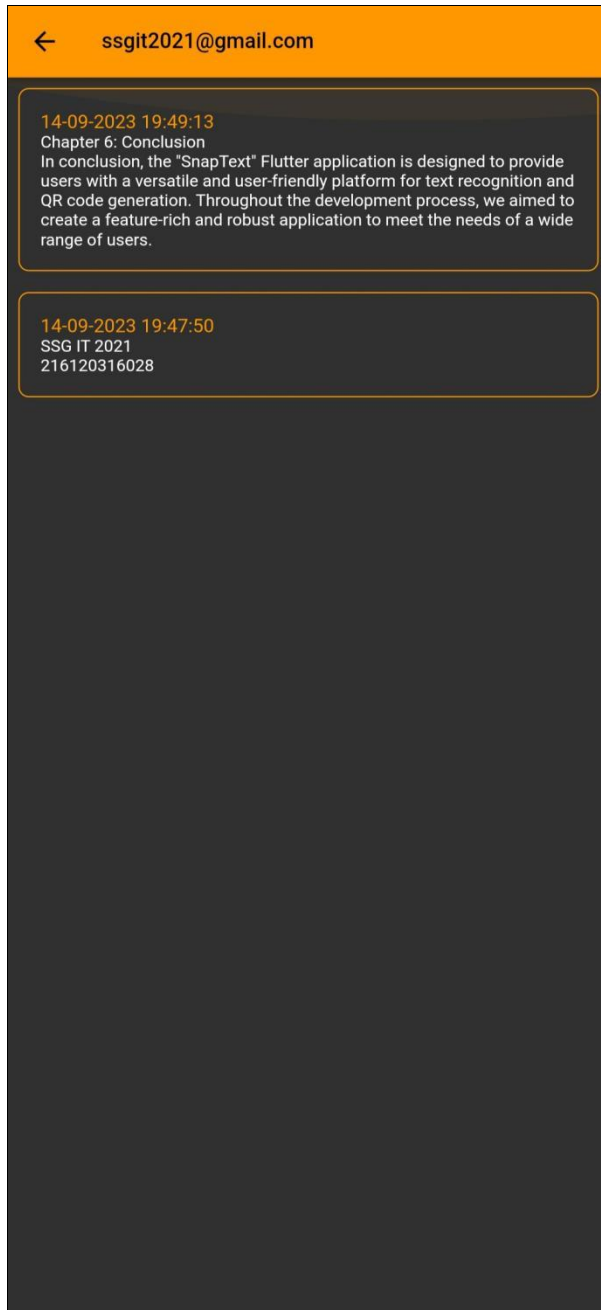
4.4 HomeScreen



4.5 HomeScreen



4.6 ResultScreen



#### 4.7 HistoryScreen

## Chapter 5: Testing

### 5.1 Login screen

| Test Case ID | Description                             | Test Steps                                                        | Expected Result                    | Pass/Fail |
|--------------|-----------------------------------------|-------------------------------------------------------------------|------------------------------------|-----------|
| TC_L01       | Successful Login with Valid Credentials | Enter valid email and password, tap "Login" button                | Redirected to Home Screen          | Pass      |
| TC_L02       | Login with Incorrect Password           | Enter valid email and incorrect password, tap "Login" button      | Error message "Incorrect password" | Pass      |
| TC_L03       | Login with Non-Registered Email         | Enter non-registered email and valid password, tap "Login" button | Error message "Email not found"    | Pass      |
| TC_L04       | Login with Empty Fields                 | Leave email and password fields empty, tap "Login" button         | Error messages for empty fields    | Pass      |
| TC_L05       | Navigate to Registration Screen         | Tap "Create Account" button                                       | Redirected to Registration Screen  | Pass      |

Fig 5.1 Testcase of Login Screen

## 5.2 Registration Screen

| Test Case ID | Description                      | Test Steps                                                                      | Expected Result                      | Pass/Fail |
|--------------|----------------------------------|---------------------------------------------------------------------------------|--------------------------------------|-----------|
| TC_R01       | Successful Registration          | Enter valid email and password, tap "Register" button                           | Redirected to Home Screen            | Pass      |
| TC_R02       | Registration with Existing Email | Enter an email that's already registered, valid password, tap "Register" button | Error message "Email already in use" | Pass      |
| TC_R03       | Registration with Weak Password  | Enter valid email and a weak password, tap "Register" button                    | Error message "Weak password"        | Pass      |
| TC_R04       | Registration with Empty Fields   | Leave email and password fields empty, tap "Register" button                    | Error messages for empty fields      | Pass      |
| TC_R05       | Navigate to Login Screen         | Tap "Already have an account? Log in" link                                      | Redirected to Login Screen           | Pass      |

Fig 5.2 Testcase of Registration Screen



## 5.3 Home Screen

| Test Case ID | Description                            | Test Steps                                      | Expected Result                    | Pass/Fail |
|--------------|----------------------------------------|-------------------------------------------------|------------------------------------|-----------|
| TC_H01       | Capture Image from Camera              | Tap "Camera" button                             | Camera app opens for image capture | Pass      |
| TC_H02       | Capture Image from Gallery             | Tap "Gallery" button                            | Gallery opens for image selection  | Pass      |
| TC_H03       | Recognize Text from Captured Image     | Capture or select an image                      | Text recognition result displayed  | Pass      |
| TC_H04       | Attempt Text Recognition Without Image | Tap "Recognize Text" without selecting an image | No action taken, no result shown   | Pass      |
| TC_H05       | Log Out                                | Tap "Log Out" button                            | Logged out and redirected to Login | Pass      |
| TC_H06       | Navigate to History Page               | Tap "History" button                            | Redirected to History Page         | Pass      |

Fig 5.3 Testcase of Home Screen

## 5.4 Result Screen

| Test Case ID | Description                   | Test Steps                                       | Expected Result                       | Pass/Fail |
|--------------|-------------------------------|--------------------------------------------------|---------------------------------------|-----------|
| TC_R01       | View Recognized Text          | Tap on a recognized text result                  | Recognized text displayed for editing | Pass      |
| TC_R02       | Edit Recognized Text          | Tap "Edit" button, modify text, and save changes | Text edited and saved successfully    | Pass      |
| TC_R03       | Search Text in Browser        | Tap "Search" button for recognized text          | Default browser opens with search     | Pass      |
| TC_R04       | Copy Text to Clipboard        | Tap "Copy" button for recognized text            | Text copied to clipboard              | Pass      |
| TC_R05       | Translate and Open in Browser | Tap "Translate" button for recognized text       | Google Translate opens in browser     | Pass      |

Fig 5.4 Testcase of Result Screen

## 5.5 History Screen

| Test Case ID | Description                          | Test Steps                           | Expected Result                                | Pass/Fail |
|--------------|--------------------------------------|--------------------------------------|------------------------------------------------|-----------|
| TC_HP01      | View History                         | Tap on a history item                | Recognized text from the selected history item | Pass      |
| TC_HP02      | Navigate to Result Page from History | Tap on a history item to view result | Redirected to the Result Page with text        | Pass      |

Fig 5.5 Testcase of History Screen

## Chapter 6: Conclusion

In conclusion, the "SnapText" Flutter application is designed to provide users with a versatile and user-friendly platform for text recognition and QR code generation. Throughout the development process, we aimed to create a feature-rich and robust application to meet the needs of a wide range of users.

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